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CHAITANYA KANDEKAR

RESEARCH PROFILE

Computational mechanics researcher specializing in phase-field fracture modeling and multi-physics coupling. Developed partitioned simulation frameworks for electro-chemo-mechanical and thermo-mechanical fracture using preCICE and finite element methods. Expertise in nonlinear FEM, solver coupling, and high-performance scientific computing with applications to stress corrosion cracking and fire-induced failure.

RESEARCH EXPERIENCE

- Jun 2021 - present **Wissenschaftlicher Mitarbeiter (Research Associate)** · Helmut-Schmidt University / University of the Federal Armed Forces Hamburg
- Project KIBIDZ - Thermo-hydro-mechanical fracture modelling and simulations for concrete under fire loads
 - Project ECoMaD - Electro-chemo-mechanical fracture modelling and simulation for stress corrosion cracking
 - Conducted tutorial lectures on continuum mechanics to master students, and MATLAB programming tutorials for bachelor students
- Apr 2019 - May 2021 **Wissenschaftlicher Hilfskraft (Research Assistant)** · Leibniz University Hannover
- Inst. of Continuum Mechanics (18 months)- Multi-scale simulations of 3 dimensional fracture using phase field in MATLAB, Nozzle simulations
 - Inst. of Dynamics and Vibration (6 months)- Fluid simulations of lubrication layer of the journal bearing in MATLAB
 - Inst. of Mechanics and Computational Mechanics (3 months)- Stochastic simulation of a beam using random field in MATLAB
- Sep 2016 - Mar 2018 **Design Engineer** · Infinarc Engineering LLP, Pune
- Mechanical Design, Calculation, 3D Modelling, Drafting, Structural and Dynamic Simulations
 - Product Development, Material Selection and Procurement, Fabrication, Assembly and Maintenance
 - Successfully designed and fabricated various machines such as Autonomous Pesticide Spraying Vehicle, Adhesive Dispensing Machine, Tube beading machine, Phosphating plant set up

EDUCATION

- Jun 2021 - present **Dr.-Ing. - Computational Engineering**
Grade: 1.0 (German scale - excellent) · Helmut-Schmidt University / University of the Federal Armed Forces Hamburg
Thesis: Multi-physics and multi-scale computational frameworks for environment-induced phase-field fracture
- Apr 2018 - May 2021 **Master - Computational Methods in Engineering**
Grade: 1.5 (German scale - very good) · Leibniz university, Hannover-Germany
Thesis: A Phase Field model for electro-chemo-mechanically induced fracture at the electrode-catalyst interface of Polymer Electrolyte Water Electrolysis cell
- Jun 2012 - Jun 2016 **Bachelor - Mechanical Engineering**
Grade: 2.2 (German scale - good) · Savitribai phule university of pune, Pune-India
Thesis: Improving Plant Capacity and productivity along with automation in filter line
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LANGUAGES

English (C1)
German (B2)
Hindi (native)
Marathi (native)

TECHNICAL SKILLS

Finite Elemental Packages - FEniCS, preCICE, Abaqus(UMAT), ANSYS
Programming Languages - MATLAB, Python, Fortran, C
CAD modeling Packages - CATIA, Solidworks
Pre/Post-processing Packages - Gmsh, Paraview, Hypermesh
Utility Packages - Microsoft Office, Latex, Github

PUBLICATIONS

- Feb 2025 *An approach for a novel multiscale partitioned framework for predicting collapse of buildings under fire loads*, 12th International Conference on Fracture Mechanics of Concrete and Concrete Structures (**FraMCoS-12**). <https://framcos.org/FraMCoS-12/Full-Papers/1319.pdf>
- Jun 2024 *Mastering the complex time-scale interaction during Stress Corrosion Cracking phenomena through an advanced coupling scheme*, Computer Methods in Applied Mechanics and Engineering (**CMAME**), Volume 428. <https://doi.org/10.1016/j.cma.2024.117101>
- April 2024 *Crack propagation in anisotropic brittle materials: From a phase-field model to a shape optimization approach*, **Engineering Fracture Mechanics**, Volume 303. <https://doi.org/10.1016/j.engfracmech.2024.110065>
- Sep 2023 *Toward coupled fire-structure simulations for forecasting smoke leakage in case of concrete under fire*, Proceedings in Applied Mathematics and Mechanics (**PAMM**), Volume 23. <https://doi.org/10.1002/pamm.202300250>
- Mar 2023 *A partitioned computational framework for damage evolution in stress corrosion cracking utilizing phase-field*, Proceedings in Applied Mathematics and Mechanics (**PAMM**), Volume 22. <https://doi.org/10.1002/pamm.202200211>
- Oct 2022 *Electro-Chemo-Mechanical Induced Fracture Modeling in Proton Exchange Membrane Water Electrolysis for Sustainable Hydrogen Production*, Computer Methods in Applied Mechanics and Engineering (**CMAME**), Volume 400, 1 October 2022. <https://doi.org/10.1016/j.cma.2022.115580>
- Feb 2017 *User Authorized Ignition System*, International Journal of Automobile Engineering Research and Development (**IJAuERD**), ISSN (print):2277-4785, VOL 7, ISSUE 1, EDITION: FEB 2017.
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CONFERENCE PRESENTATIONS

- Sep 2025 *Computational framework for coupled simulations of concrete structures under fire loads*, Baustatik und Baupraxis Forschungskolloquium (**IBB 2025**), Bad Ball, Germany
- Sep 2025 *An approach for a novel multiscale partitioned framework for predicting collapse of buildings under fire loads*, ECCOMAS International Conference on Computational Plasticity (**COMPLAS 2025**), Barcelona, Spain
- July 2025 *A partitioned computational framework for exploiting damage in stress corrosion cracking of metal matrix composite using phase-field method*, International Conference on Damage Mechanics (**ICDM 2025**), Singapore, Singapore
- July 2025 *On a hybrid simulation using a phase-field approach for predicting the concrete behavior under fire*, European Solid Mechanics Conference (**ESMC 2025**), Lyon, France
- Sep 2024 *Film rupture-dissolution-repassivation phase-field model to exploit damage evolution in stress corrosion cracking: a partitioned computational framework*, Material Science and Engineering (**MSE 2024**), Darmstadt, Germany
- Sep 2023 *An advanced multi-phase-field approach for coupled electro-chemo-mechanical problems using the preCICE library applied to fcc microstructure*, ECCOMAS International Conference on Computational Plasticity (**COMPLAS 2023**), Barcelona, Spain
- Aug 2023 *An advanced multi-phase-field approach for electro-chemo-mechanical dissolution driven SCC*, European Corrosion conference (**EUROCORR 2023**), Brussels, Belgium
- Jun 2023 *An advanced coupling approach for Solving Corrosion and Fracture Mechanics using preCICE*, ECCOMAS International conference for Coupled Problems (**COUPLED 2023**), Chania, Greece
- Aug 2022 *Multi-phase-field framework for damage evolution in stress corrosion cracking*, European Corrosion conference (**EUROCORR 2022**), Berlin, Germany
- Aug 2022 *Computational framework for electrochemical dissolution driven stress corrosion cracking utilising a multi-phase-field approach*, 92nd International Association of Applied Mathematics and Mechanics (**GAMM 2022**), Aachen, Germany
- Feb 2022 *A coupled multifield modelling approach to describe material behaviour of concrete under fire load*, 147th Nord-deutsche Mechanik Kolloquium (**NDMK 2022**), Braunschweig, Germany
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EXTRA-CURRICULAR PROJECTS

- Sep 2018- Sep 2019 **Formula Student Competitions 2019** · FSN, FSA, FSS
- Team - 'Horsepower Hannover' participating for Netherlands, Austria and Spain
 - International competitions for designing and building formula prototype electric cars
 - Working in the chassis department for designing and testing Carbon Fibre sandwiches for mono-coque chassis and also responsible for Structural Equivalency Spreadsheet
- Sep 2014- Jul 2015 **Formula Student Competitions 2015** · SAE SUPRA
- Team - 'The Metal Falcons' participated for national competition in India
 - All India Rank – 9th out of 167 teams
 - Led the team as a Steering system department and Innovation department head. Worked in the design, analysis and fabrication of steering system. Presented the 'Cost Analysis Report'
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HONOURS & AWARDS

- Jul 2015- May 2016 Elected as the student council member – '**Technical Secretary**' and represented 6000 students, the university, and carried out 26 technical activities and events
- Dec 2015 **Campus Ambassador** - represented Engineering Cluster Pune and D Y Patil College, Pune
- Dec 2014 **Campus Ambassador** for SGT International- organised 'Auto Factory Bikers Workshop' by IIT-BHU